AI-GNITION 2026 · PROBLEM STATEMENT 1 · TEAM JARDS

Nest Helper

Smart size, perfect time, zero stress.

Project: Nest Helper, a personalised retention chatbot for Nestling (a fictional Singapore baby shop) · NPS International School Singapore · Jeiyanth, Devanshi, Adhvik, Shivangi, Reyansh

The problem

Babies change on a schedule: 5 diaper sizes and 6 clothing sizes in 18 months, and a new diaper pack every 7 to 12 days. Many shops send everyone the same promotion, so families can run out, outgrow sizes or drift away. Singapore also restricts infant formula promotion (SIFECS).

Target users: parents of babies from newborn to 3 years in Singapore, and the small baby shops that want them to come back.

Our solution

A MantaGO chatbot on the shop website, driven by a decision engine. It reads each family's orders, baby age and consent, runs safety checks, then picks **one** message: refill, size-up, milestone or win-back. Each message is written for that baby and explains its data on request.

How one message is chosen

1 Consent
Agreed to reminders? If not, stay quiet.

2 Formula lock
Only infant formula due? Stay quiet.

3 Cooldown
Snoozed or messaged in 7 days? Wait.

4 Journey
Refill, then size-up, milestone, win-back.

5 Message
Reorder, wait, stop, "Why?", or a human.

13 of 20
fictional families messaged

7 of 20
deliberately held back

27
message packs in the live bot

36 of 37
tests pass (1 not run)

Working now

- Website, 26 product pages, cart and tracked links
- Chatbot: consent, reminders, "Why this message?", reorder, STOP, human handover
- AI answers from a shop-facts file, with a privacy rule
- Decision engine, message checker (13 of 13 real messages pass, 7 of 7 bad ones blocked)
- Testing caught 11 problems, all fixed, including the AI reading out a phone number

Planned, not built

- Run the engine daily and send each pack as a scheduled broadcast (needs a MantaGO plan with scheduling)
- A real "Remind me in 3 days" snooze
- The bot calling the engine's API directly (built and tested locally)
- Reading click data to measure results

Responsible by design

- Messages only to families who agreed; STOP at any time; one message per 7 days
- No offers, bundles or bot-started messages for infant formula; no brand or feeding advice
- The bot never shares customer details
- Nest Helper tells parents it is an AI assistant

How impact would be measured

- **Repeat purchase:** families who reorder within 30 days of a reminder ÷ families reminded
- **Retention:** families with an order in the last 90 days ÷ all families
- **Lifetime value:** revenue per family over 12 months
- Each compared with a 20% holdout who get no reminders. No results are claimed.

Tools and technology

MantaGO chatbot with Live Chat handover; its AI answers use Google's Gemini 3.1 Flash-Lite, from our shop-facts file · **Python 3** engine and message checker (optional FastAPI API) · **GitHub Pages** website with UTM-tagged links · Excel test log. Our team used an AI assistant to help draft parts of the code and text.

Real vs simulated: the bot, engine, website and tests are real. The shop, customers, orders, prices and all metric results are simulated.
Try it: jefire916.github.io/JARDS_PS1_Nestling (tap the chat bubble, type DEMO) · **Code:** github.com/Jefire916/JARDS_PS1_Nestling · **Bot ID:** olejqMRdj